EnergyWise UK

AI-powered energy intelligence for every UK household

Venture Spark Programme 2026

Presented by Ayan Dhar

The Problem

80%

of UK households overpay on energy bills

£300+

average annual overspend per household

29M

UK households lack energy visibility

UK energy bills have risen 54% since 2021. Yet consumers still receive paper bills with no actionable insight. Most have no idea if they are on the cheapest tariff, how their usage compares, or what their carbon footprint looks like. The problem is not energy cost alone — it is the total absence of accessible, personalised energy intelligence.

"I have no idea if I'm on a good deal — my bill is just numbers." — UK Consumer Survey, 2025

The Solution

EnergyWise is an AI-powered energy dashboard that scans your bills, compares tariffs, tracks carbon footprint, and delivers personalised savings advice — all in one place.

AI Bill Scanner

Upload a bill photo and Gemini AI extracts provider, tariff, usage & cost data instantly.

Usage Analytics

Interactive charts track electricity and gas consumption trends over time.

EPC & Carbon

Postcode-based EPC rating lookup and real-time CO_2 emissions tracking.

Tariff Comparison

Side-by-side comparison of current vs available tariffs to find savings.

Live Demo: venture-spark-uk.lovable.app

Market Potential

TAM

£8.7B

UK household energy management market

SAM

£1.2B

Digitally-engaged UK households seeking energy tools

SOM

£24M

Year 3 target: 80K subscribers at £25/mo

Sources: Ofgem, Energy UK, Statista 2025. SOM based on 0.3% penetration of digitally active households.

Unique Value Proposition

"Unlike price-comparison sites that give you a one-off quote, EnergyWise gives you an ongoing, AI-powered energy command centre."

01

AI-First Intelligence

Gemini-powered bill scanning extracts data from photos — no manual entry, no OCR errors.

02

Holistic Dashboard

Usage tracking, tariff comparison, EPC ratings, carbon footprint, and budget tools in one platform.

03

Personalised & Persistent

Learns your household profile over time. Every bill builds a richer picture of your energy story.

£ Business Model

Free Tier

£0/mo

- Manual bill entry
- Basic dashboard
- Savings tips

Smart

£9.99/mo

- AI bill scanning
- Tariff comparison
- Carbon tracking

Pro

£24.99/mo

- Unlimited scans
- Household profiles
- Priority support
- API access

Revenue mix: 70% subscriptions, 20% affiliate commissions from energy switching, 10% data insights (anonymised).

Go-to-Market Strategy

Content & SEO

Energy-saving guides and postcode comparison tools driving organic traffic. Target: 50K monthly visitors within 12 months.

Community Partnerships

Partner with housing associations, local councils, and fuel poverty charities to reach underserved households.

Referral Engine

Users who save money become advocates. In-app referral rewards: 1 month free for every 3 referrals.

Conversion funnel: Free sign-up → AI scan first bill → show savings → convert to paid plan (target: 12% free-to-paid).

🚀 MVP & Traction

What we built this sprint:

- Full-stack energy dashboard with auth, profiles, and bill history
- AI bill scanner using Gemini 2.5 Flash via secure edge function (JWT-protected)
- Interactive usage charts, budget tracker, and tariff comparison engine
- EPC rating lookup by postcode and real-time carbon footprint calculator
- Dark/light mode, responsive design, and production deployment on Lovable Cloud

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Riskiest assumptions tested:

Can AI reliably extract structured data from energy bill photos?

Yes — Gemini 2.5 Flash achieves 90%+ accuracy on UK bill formats.

Will users trust an automated tool with their bill data?

JWT auth + RLS policies ensure data isolation; early testers reported high confidence.

The Team

Ayan Dhar

Founder & Developer | EnergyWise UK

MSc Data Science & Analytics

Cardiff University

4+ years as a Cloud Data Engineer

Expertise in cloud infrastructure, data pipelines, and scalable backend systems — the backbone of EnergyWise's AI-powered bill analysis and real-time energy insights.

BEng Electronics & Communication Engineering

Strong foundation in systems thinking, signal processing, and hardware-software integration.

Why me: I bring the rare combination of cloud engineering depth, data science rigour, and first-hand understanding of UK energy costs as a student and consumer.

Budget Breakdown

Seed round: £150,000 over 18 months

Category	Amount	% of Total	Purpose
Product & Engineering	£65,000	43%	2 full-time developers, AI/ML costs, infrastructure
User Acquisition	£35,000	23%	Content marketing, SEO, community partnerships
Operations	£25,000	17%	Cloud hosting, Supabase, API costs, legal
Team & Hiring	£15,000	10%	Part-time energy advisor, customer support
Contingency	£10,000	7%	Buffer for market shifts and unexpected costs

Runway: 18 months to Series A metrics (10K paid subscribers, £120K ARR).

Impact & Scalability

2.4M tonnes	potential CO_2 reduction if 10% of UK households optimise energy use
£8.7B	annual household savings unlocked through smarter tariff switching
29M	UK households addressable — expanding to EU energy markets in Year 3

Network Effects & Scale:

More users = richer anonymised benchmarking data = better recommendations = higher retention. Each new bill scanned improves AI accuracy for all users.

→ Next Steps

Top 3 actions — next 30 days:

01 **Launch Beta**
Open to 500 beta users via waitlist. Measure activation, retention, and free-to-paid conversion.

02 **Partner Outreach**
Secure 3 housing association partnerships for distribution to 10,000+ households.

03 **Investor Conversations**
Close seed round commitments. Target: £150K from angels and climate-tech funds.

What unlocks the next 6 months:
Seed funding + 1,000 active users + 2 energy provider API integrations

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